
Exercise 1 – Create and Read a File

Workflow

```
create_file
  ↓
read_file
```

Task 1

Create:

```
/tmp/message.txt
```

Content:

```
Welcome to Apache Airflow
Learning DAGs
Learning Task Dependencies
```

Task 2

Read the file and print its contents in task logs.

Exercise 2 – Employee Salary Report

Workflow

```
create_salary_file
  ↓
calculate_total_salary
```

↓
generate_report

Input File

/tmp/employees.txt

Content:

```
Rahul,45000
Priya,52000
Amit,61000
Sneha,48000
```

Task 2

Calculate total salary.

Expected:

```
Total Salary = 206000
```

Task 3

Create:

/tmp/report.txt

Content:

```
Salary Report
Employees = 4
Total Salary = 206000
```

Exercise 3 – Student Marks Processing

Workflow

```
create_marks_file
```

↓

```
calculate_average
```

↓

```
generate_result
```

Input

```
/tmp/marks.txt
```

Content:

```
Math,80
Science,75
English,90
Python,95
```

Task 2

Calculate average marks.

Expected:

```
Average = 85
```

Task 3

Generate:

```
/tmp/result.txt
```

Content:

```
Average Marks = 85
Result = PASS
```

Exercise 4 – Product Stock Alert

Workflow

```
create_inventory
  ↓
find_low_stock
  ↓
generate_alert
```

Input

```
Rice, 50
Oil, 7
Soap, 35
Sugar, 10
Tea, 5
```

Low Stock Rule

Stock < 15

Output File

```
/tmp/alerts.txt
```

Expected:

```
Oil
Sugar
Tea
```

Exercise 5 – Attendance Report

Workflow

```
create_attendance
    ↓
count_present
    ↓
count_absent
    ↓
generate_summary
```

Input

```
Rahul, Present
Priya, Present
Amit, Absent
Sneha, Present
Kiran, Absent
```

Output

```
Total Students = 5
Present = 3
Absent = 2
```

Generate:

```
/tmp/attendance_report.txt
```

Exercise 6 – CSV Processing

Workflow

```
create_csv
  ↓
read_csv
  ↓
calculate_revenue
  ↓
create_summary
```

sales.csv

```
product,quantity,price
Laptop,2,70000
Mouse,5,500
Keyboard,3,1200
```

Calculate Revenue

Expected:

```
Laptop = 140000
Mouse = 2500
Keyboard = 3600

Total Revenue = 146100
```
